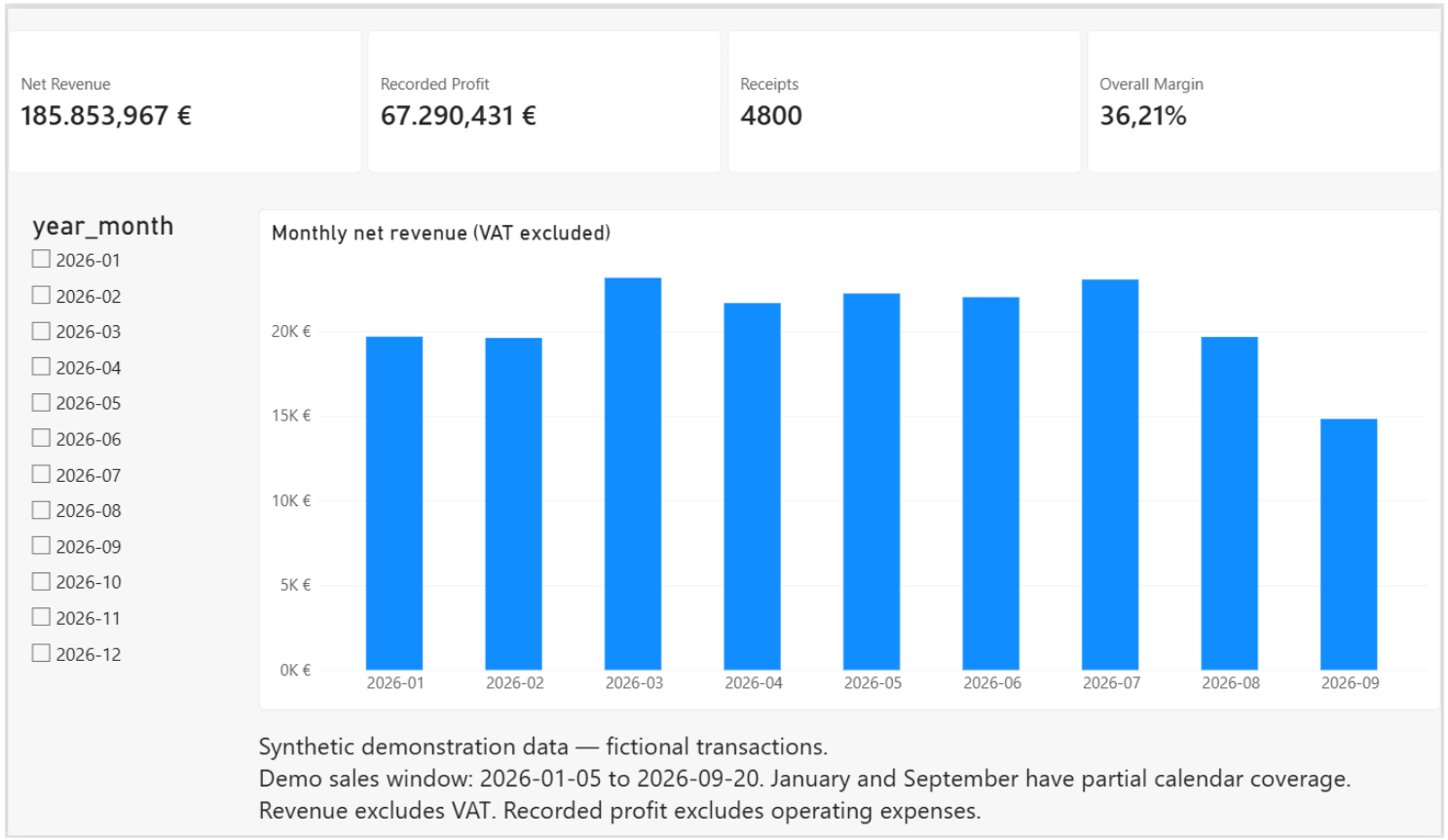
PharmaBI

Retail analytics portfolio

Danilo Caporicci

R, SQL and SQLite
Shiny and Power BI



Power BI sales overview

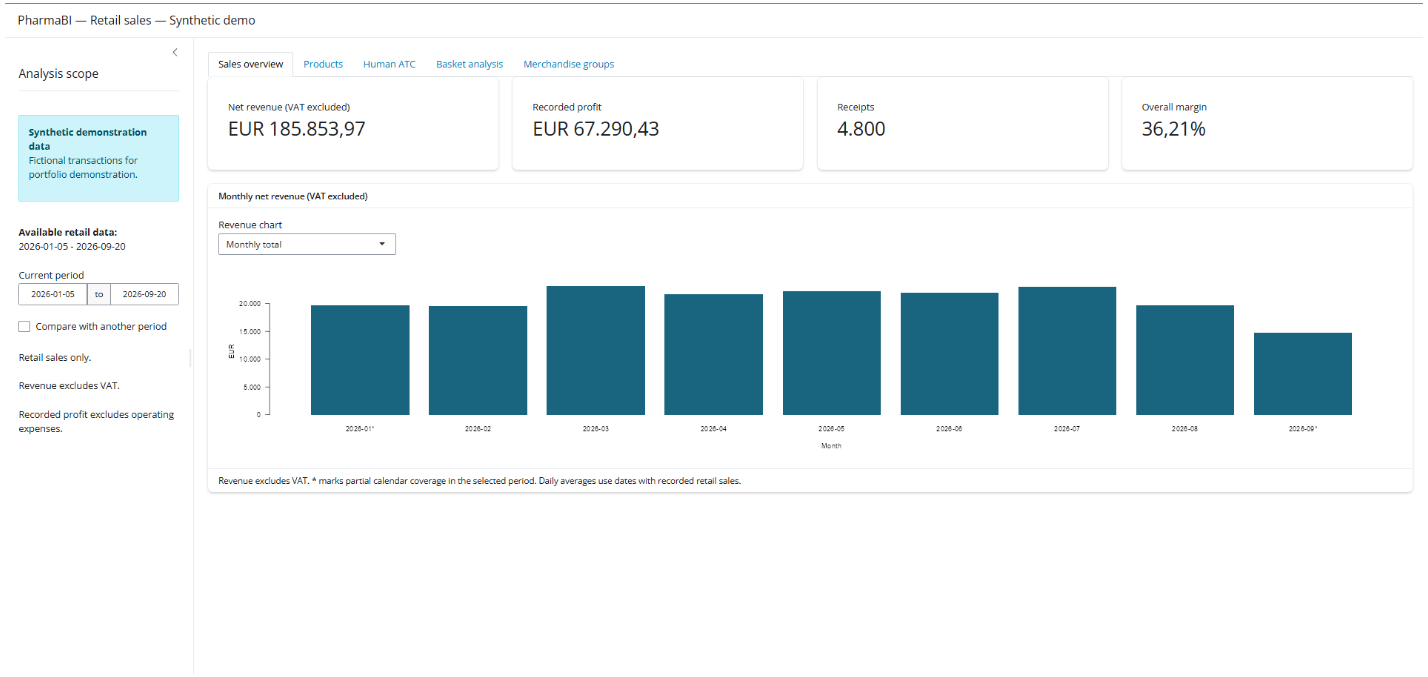
Model, validation and dashboards

Data grain

One row per receipt and product

70

automated checks passed
on Windows 11 and Linux Mint



4.800 retail receipts
240 fictional products

RDS, SQLite and CSV totals reconcile

Shiny for exploration. Power BI for reporting.

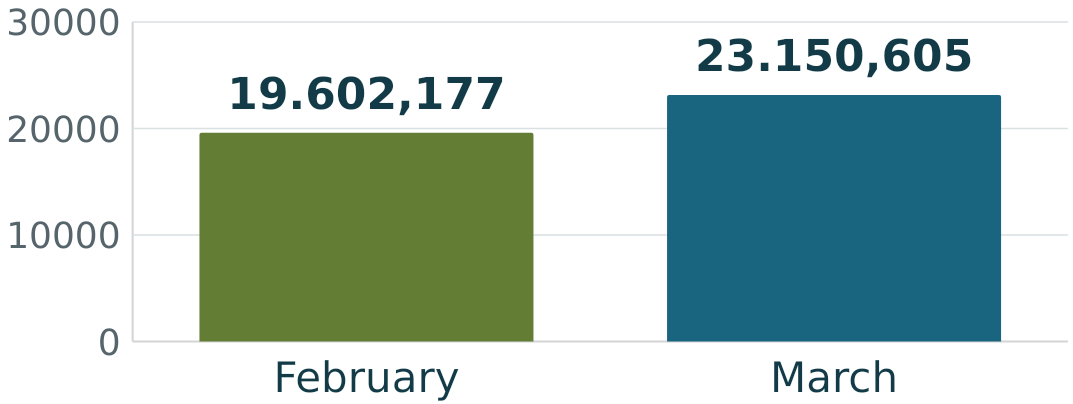
Monthly totals and daily averages

March 2026 compared with February 2026

Net revenue

+18,10%

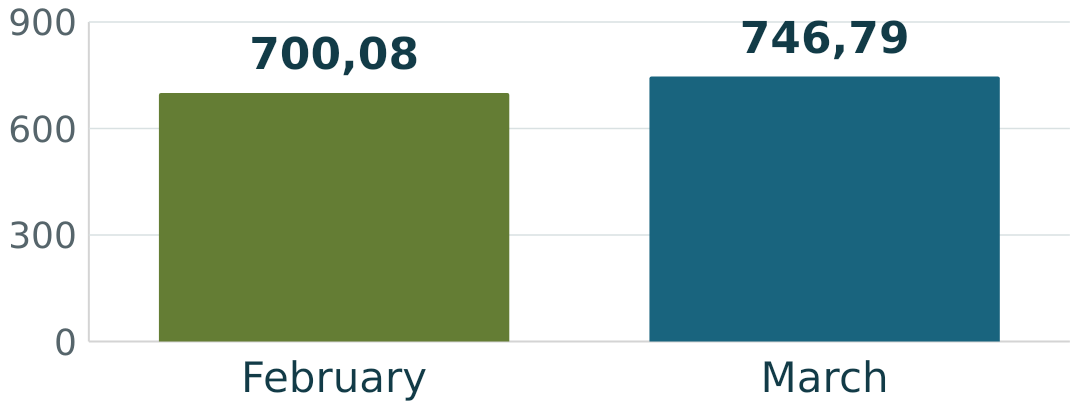
EUR



Net revenue per active day

+6,67%

EUR per active day



Active days: 28 in February and 31 in March

Net revenue excludes VAT. Active days are dates with recorded retail sales.

Synthetic demonstration data

Volume, profit and margin

Different metrics identify different product leaders

Product	Net packages	Profit (EUR)	Margin
Supplements 042 D00000042	180	2.555,283	50,47%
Skin care 061 D00000061	220	422,180	21,22%
Other non-prescription medicines 024 D00000024	20	42,900	54,75%

Highest profit: Supplements 042. Highest volume: Skin care 061.

Margin ranking: at least 10 net packages, positive revenue and recorded costs above zero.

Frequency, confidence and lift

Product pairs across 4.800 retail receipts

Codes (suffixes)	Shared receipts	Support	Confidence A to B	Lift
D41 + D42	133	2,7708%	80,12%	25,81
D101 + D102	129	2,6875%	92,14%	29,88
D61 + D62	119	2,4792%	68,00%	22,05
D45 + D103	1	0,0208%	20,00%	192,00

Lift 192,00 rests on one shared receipt

An exploratory threshold of 5 shared receipts retains the first three pairs.

The generator deliberately includes these three repeated associations.

Scope and next steps

Current scope

Sales performance
Product and category rankings
Receipt structure and associations

Further work

Purchases and stock data
Longer sales history
Deployment and wider testing



Recorded profit excludes operating expenses.
ATC categories describe products, not diagnoses.

github.com/dcaporicci/PharmaBI-portfolio